projectsf — Investment Analysis Toolkit

A guide to the three projects: what they are, how to run them, what they produce, and what it all means.
Author: Ali Saleh Al Taher · Built with Python & Excel · Research artifact, not investment advice.

1 What this is (the one-paragraph version)

Three self-contained tools that apply an engineer's discipline to investing: pull real data, build the model, test the assumptions, then take a view. Project 1 values *any* US-listed stock from live filings. Project 2 maps the AI-infrastructure supply chain to find where it is physically constrained. Project 3 underwrites a real-estate deal the way an institutional sponsor does, splitting the return between investors and the manager. Every number is either sourced from official filings or clearly flagged as an assumption.

2 The three projects

1 — Equity Valuation projectsf\equity-valuation

Type a ticker; it pulls audited financials from the U.S. SEC (EDGAR) and a live share price, then builds a full valuation: a discounted-cash-flow (DCF) model with a cost-of-capital (WACC) build, a Bear/Base/Bull scenario band, a comparable-company cross-check, a Monte-Carlo simulation, and a one-click **automated research note** that scores the company on globally recognised techniques.

Technique	What it answers
DCF + WACC	What are the future cash flows worth today?
DuPont ROE	Is the return driven by margins, efficiency, or leverage?
ROIC vs WACC	Does the business actually create value?
Piotroski F-Score	9-point fundamental strength check
Altman Z-Score	Bankruptcy / financial-distress risk
Beneish M-Score	Are the earnings possibly being manipulated?

Sloan accruals	Are earnings backed by real cash?
Reverse DCF	What growth is the current price already assuming?

2 — AI Bottleneck Map projectsf\ai-bottleneck-map

Scores each layer of the AI hardware chain (GPUs → high-bandwidth memory → advanced packaging → power & grid → copper & uranium) on a transparent **"tightness"** metric — utilisation, lead time, supply inelasticity, supplier concentration — to identify the *binding constraint* (currently advanced packaging / CoWoS) and show where investment capital is migrating. The thesis: own the bottleneck, not the headline chip.

3 — Real-Estate ROI projectsf\real-estate-roi

Underwrites a value-add property deal (buy → renovate → grow income → sell) and runs an institutional **LP/GP equity waterfall**: return of capital → preferred return → promote tiers. It reports project IRR, the investor (LP) IRR after the manager's cut, equity multiple, cash-on-cash, DSCR and debt yield — the exact mechanics Blackstone/BlackRock-style real-estate funds use.

3 How to run them

Easiest — double-click (no typing):

- Open `C:\Users\ali12\projectsf` in File Explorer, open a project folder.
- Double-click `setup.bat` once (installs dependencies). Wait for "Done", close it.
- Double-click `run.bat` to run, or `app.bat` for the visual web app.

From PowerShell (these call Anaconda's Python directly, so nothing needs to be on your PATH; PowerShell doesn't accept `&&`, so run one line at a time):

```
cd $env:USERPROFILE\projectsf\equity-valuation
& "$env:USERPROFILE\anaconda3\python.exe" -m pip install -r requirements.txt
& "$env:USERPROFILE\anaconda3\python.exe" value.py MU      # valuation of any ticker
& "$env:USERPROFILE\anaconda3\python.exe" analyze.py MU    # full research note
& "$env:USERPROFILE\anaconda3\python.exe" -m streamlit run app.py   # web app
```

Projects 2 & 3 are the same pattern: `build.py` (bottleneck) and `roi.py` (real estate). The web app for any project is `app.bat`.

4 What outputs each model produces

Outputs are **regenerated every time you run** and land in each project's `output\` folder.

Model	Outputs & what they mean
Equity Valuation value.py	`<TICKER>_valuation.xlsx` — the full DCF model with live formulas · `_valuation.json` — the same results as data · `_montecarlo.png` — the fair-value distribution chart · plus a console summary.
Research note analyze.py	`<TICKER>_research.md` — a written analyst note (reads like a one-pager) · `_analysis.xlsx` — every ratio/score in one workbook.
MU thesis mu_thesis_model.py	`MU_thesis.json` — the Bear/Base/Bull scenario fair values + the written memo in `THESIS_MU.md`.
Bottleneck Map build.py	`bottleneck.json` — the scored layers (data the web app draws) · a console ranking · the visual lives in the `app.bat` web app.
Real-Estate ROI roi.py	`deal_underwriting.xlsx` — underwriting + the LP/GP waterfall with live formulas · `<deal>_roi.json` — all results as data · console summary.

> **Yes — that is the nice part:** the Excel files aren't dead screenshots. They contain the *live equations* — DCF discounting, WACC, IRR, the waterfall — all linked to a blue-free Assumptions tab. Change an input cell and Excel recalculates the whole model in real time. The Python builds the workbook; Excel lets you stress it by hand. (The numbers are verified to match the Python engine under a real recalculation.)

5 The Micron (MU) analysis — what it means & what I did

What I did. Rather than let the tool auto-pick assumptions, I built a deliberate, hand-made model of Micron through the memory cycle: three scenarios (Bear / Base / Bull), each with an explicit revenue-growth path, margin path, capex assumption and discount rate — every driver chosen and footnoted in `THESIS_MU.md`. The financials underneath are real (audited SEC filings); the judgement is mine and written down.

What it means. Micron is the cleanest listed way to own *high-bandwidth memory (HBM)* — one of the tightest links in the AI chain (it connects directly to Project 2). The model says this is *not* a cheap stock on a normalised mid-cycle; it is an **asymmetric bet on HBM staying scarce**:

Scenario	Fair value / share	It hinges on
Bear (25%)	~$12	Memory oversupply returns; margins collapse
Base (50%)	~$60	HBM tight into 2026, DRAM cycle normalises
Bull (25%)	~$127	HBM stays the binding constraint; structurally higher margins

The recommendation that falls out: **accumulate on weakness, size for the cycle** — a conviction position, not a core holding, re-underwritten at every capex update. The three swing factors are HBM share/pricing, where mid-cycle margins settle, and capex discipline. It would be *wrong* if Samsung qualifies HBM at scale or hyperscaler AI spending digests. (These fair values are independent of the live share price, so they hold regardless of any given day's quote.)

6 Honesty & limitations

- Fundamentals are audited 10-K values from SEC EDGAR; anything a filing doesn't provide is flagged, never invented.
- Scenario drivers and peer multiples are explicit, editable assumptions — the models are decision tools, not oracles.
- Coverage is US-listed companies (EDGAR). Prices come from a free public feed; always sanity-check the quote.
- This is a research / portfolio artifact and **not investment advice.**

Read alongside: `INVESTMENT_THESIS.md` (the through-line), and each project's `README.md` and `THESIS.md`.